31/10/26

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Assignment-2

1) TASK 1: linear regression using Ordinary least squares:

Given, $x = 1,2,3,4,5$ & $y = 2.1, 3.9, 6.2, 7.8, 10.1$

33/50

Using the normal equation

$\Sigma x = 15$, $\Sigma y = 30.1$, $\Sigma x^2 = 55$, $\Sigma xy = 110.2$

Mean: $\bar{x} = 3$ & $\bar{y} = 6.02$

slope: $w_1 = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2 = 19.9/10 = 1.99$

Intercept: $w_0 = \bar{y} - w_1 \bar{x} = 6.02 - 1.99(3) = 0.05$

$y = 0.05 + 1.99x$

At $x = 6$: $\hat{y} = 0.05 + 1.99(6)$

$\Rightarrow 11.99$

2) TASK 2: Ridge Regression:

Given: $S_{xx} = 10$, $S_{xy} = 19.9$ $\lambda = 5$

least-square slope $= S_{xy} / S_{xx} = 19.9/10 = 1.99$

$\Rightarrow 19.9/(10+5) = 1.3267$

Percentage shrinkage

$\Rightarrow [(1.99 - 1.3267)/1.99] \times 100$

$= 33.33\%$

3) TASK 3: Naive Bayes classification:

From the 10 emails: Spam = 5 & Ham = 5.

Therefore, P(Spam) = P(Ham) = 5/10 = 0.5

For spam, P(offer = Y/spam) = 4/5 = 0.8

P(link = Y/spam) = 4/5 = 0.8

For Ham, both probabilities are 1/5 = 0.2.

Spam score = 0.5 × 0.8 × 0.8 = 0.32

Ham score = 0.5 × 0.2 × 0.2 = 0.02

Y Probability = [0.5(5/13)^2] / [0.5(5/13)^2 + 0.5(2/13)^2]

= 25/29

≈ 0.8621

4) TASK 4: One batch Gradient Descent step:

Initially $w_1 = w_2 = 0$

$\rightarrow -\ln(0.5) = 0.6931$

$\partial L/\partial w_1 = [(0.5 - 1)(1) + (0.5 - \sigma(2))(1)]/2 = 0.25$

$\partial L/\partial w_2 = [(0.5 - 1)2 + (0.5 - \sigma(1))(1)]/2 = -0.25$

For (1, 2), z = 0.125, $p = \sigma(0.125)$

Mean loss update = $-\ln(0.5312)$

≈ 0.6362

TASK 5: Entropy, Information Gain & Gini Impurity.

Parent node: 10 Samples (6 Yes, 4 No)

Entropy (parent) = $-0.6 \log_2 0.6 - 0.4 \log_2 0.4$

= 0.5510

Information gain = 0.971 - 0.5510

= 0.420 bits

Gini (Parent) = $1 - 0.6^2 - 0.4^2$

= 0.48

Gini (q_2) = $1 - (1/2)^2 - (2/2)^2 = 0.4444$

Gini decreases = 0.48 - 0.44

= 0.2133

6) TASK 6: Cost Complexity pruning:

Sub tree = training errors + α (no. of leaves)

= 3 + 4α

Single leaf cost = 8 + α

At equal cost = 3 + 4α = 8 + α

For α = 0.01: subtree cost = 3.04

leaf cost = 8.01

For α = 0.02: subtree cost = 3.01

leaf cost = 8.02

7) Task 7: ANN

Given, $x = (1, 2)$, target $w_1 = 0.4, w_2 = 0.3, v = 0.5$

Sigmoid $= 1/(1+e^{-z})$

Hidden net $= 0.4(1) + 0.3(2) = 1$

$h(\sigma) = 0.7311$

Output net $= 0.5(0.7311) = 0.3655$

$\hat{y} = \sigma(0.3655) = 0.5904$

Initial loss:

$E = \frac{1}{2}(1 - 0.5904)^2 = 0.0839$

8) Task 8: Support Vector Machine (SVM):

Given $w = (0.5, 0.5)$,

$b = -2$

$f(x) = 0.5x_1 + 0.5x_2 - 2$

norm $(\|w\|) = \sqrt{(0.5)^2 + (0.5)^2} = 0.7071$

For $(3,3)$, $f = 1$ for $(1,1)$, $f = -1$

Decision boundary: $x_1 + x_2 = 4$

For $(4,2)$, $f = 1$

distance $= 1/\|0.7071\| = 1.414$

For $(1,2)$, $y = -1$ & $f = 0.5$ loss $= 1$

For $(2,3)$ $= y = +1$ & $f = 0$; loss $= 1$

Task 9: Ensemble learning majority

Each classifier, has accuracy 1-0.70

error = $q = 0.30$

For 3 classifiers, majority accuracy = $C(3,2)(0.3)^2(0.3)+(0.3)^3$

$= 0.441 + 0.343$

$= 0.784$

$= 78\%$

For 5 classifiers = $C(5,3)(0.3)^3(0.2)^2 + C(5,4)(0.3)^4(0.3) + (0.3)^5$

$= 0.3087 + 0.26015 + 0.16807$

$= 83.69\%$

Task 10: AdaBoost weight update

Each weight = 1/5 = 0.2

Two samples (2 & 5)

weighted error ε = 0.2 + 0.2

= 0.4

$\alpha = 1/2 \ln[(1-\varepsilon)/\varepsilon]$

$= 1/2 \ln(0.6/0.4) = 0.2027$

Sample = $0.2 e^{(-\alpha)} = 0.1633$

Sum = $3(0.1633) + 2(0.2449)$

= 0.4798.

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1d) Task 11 : Linear Discriminant

Given, $\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 2 \end{bmatrix}$

$\mu_1 = (1,2)^T$ & $\mu_2 = (3,1)^T$

$\mu_2 - \mu_1 = (2,-1)^T$

$\Sigma^{-1} = (1/1.75)\begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}$

Therefore $w = \Sigma^{-1}(\mu_2 - \mu_1) = (10/7, -12/7)^T$

$= 1/2(\mu_1 + \mu_2)^T w = 1/2(4/7) : (10/7, -12/7) = 2/7 = 0.285\overline{7}$

$x = (2.5, 1)$

$w^T x = (10/7)(2.5) - (12/7)(1) = 13/14$

$= 0.9286$

12) Task 12 : Classifier Evaluation & Cost-Based Selection

Formulas : Accuracy $= (TP + TN)/1000$

Precision $= TP/(TP + FP)$

Recall $= TP/(TP + FN)$; F-1 $= 2PR/(P + R)$

Model A $\Rightarrow$ Accuracy $= (70 + 810)/1000$

Model B $\Rightarrow$ Accuracy $= (85 + 720)/1000$

Cost $= FN \times ₹50,000 + FP \times ₹5000$

A : $20(50,000) = ₹16,50,200$

$\Rightarrow$ 85%.